

RESULTS, METHODS, AND DEVIATIONS

Advisor - Prof. Vlad Veksler

Researcher - Peter Kiprop

Methods and Deviations

This report details differences between the executed study and the original proposal and justifies each deviation. All deviations were necessary because commercial model APIs changed after the proposal was written. No discretionary changes were made to the research design. The report also identifies which design elements remain unchanged, thus confirming that the study continues to address its original research question.

Summary of what changed

Element	As proposed	As executed	Cause
Test model 1	Claude-3-5-sonnet	claude-sonnet-5	Provider retirement
Test model 2	gpt-4o	gpt-5.5	Migrated for generational parity.
Evaluator	gemini-1.5-pro	gemini-3.7-flash	Provider retirement, then access restriction.

Evaluator count	One judge	Two judges (primary + rotating secondary)	Design strengthening.
Temperature	0.2, both models	Provider defaults	Parameter removed or pinned by providers.
Generation max_tokens	2000	4000	Truncation risk to structured output.
Judge max_tokens	1024	4000	Systematic evaluation failures on complex inputs.

Model Substitutions

Both models specified in the proposal were retired by their providers prior to data collection. Anthropic discontinued claude-3-5-sonnet on the API, and gpt-4o, although still available, is from 2024. To prevent a significant capability gap and confounding in cross-model comparisons, both test models were updated simultaneously. The replacements, claude-sonnet-5 and gpt-5.5, are comparable in tier and release date.

The evaluator model was also withdrawn. Gemini-1.5-pro is no longer available, and its intended replacement, gemini-2.5-pro, is listed by the API but returns a 404 error for new users. Access appears restricted to existing keys, and the project's key cannot access any Gemini 2.5 model,

either Pro or Flash. Among the remaining callable non-preview options, all are Flash-tier; therefore, I used a Flash-tier judge instead of a Pro-tier judge. This substitution is justified because the scoring task involves comprehension against an explicit constraint list rather than advanced reasoning, and because family neutrality, the ability of a judge to score outputs from both test models without evaluating its own family, is the critical property underpinning the judge design. Shifting the primary judge away from Google to maintain tier would have required scoring every output with a model from one of the two test families, which would have introduced greater bias.

Claude Fable 5 and Claude Mythos 5 were both considered and rejected. Mythos is not publicly available, since access is by invitation only through Anthropic’s Project Glasswing. Fable 5 is generally available but ships with safeguards that route a subset of queries to Claude Opus 5 instead, which disqualifies it for an evaluator because the model that answers is not guaranteed to be the model selected on an unknown and unidentifiable subset of calls. A judge whose identity is not knowable per call cannot support the reliability claims this design rests on. Both are also Anthropic models, so either would have collided with Claude-sonnet-5 under the family-neutrality rule anyway.

Decoding Parameters

The proposal specified a temperature setting of 0.2 for both models to reduce sampling variance and enhance representativeness. This configuration is no longer feasible. Claude-sonnet-5 does not accept the temperature parameter and instead provides a discrete effort control, while gpt-5.5 accepts the temperature parameter but only at the default value of 1.0. Consequently, both

models operate at their provider defaults, recorded as null for Claude and 1.0 for GPT, accurately reflecting their configurations. For OpenAI calls, reasoning effort is explicitly set to low to ensure the study record reflects the requested setting rather than an unobservable provider default. Claude's effort setting remains unset, as the two controls are not directly comparable and equating them would imply an unjustified equivalence.

This change does not impact the primary comparison between baseline and CRAFT conditions within each model. Both conditions within each pair use identical decoding settings, enforced by a settings hash that rejects any pair with differing parameters. The comparison remains valid because settings are constant within each pair, not because of their absolute values. The change increases run-to-run variance, particularly for GPT at temperature 1.0, which elevates the stability subset beyond its original role as a confirmatory check and makes it the sole empirical measure of variance.

What was retained unchanged

The central question is whether the study still addresses its original research objective, which it does. The task set remains unchanged, comprising 50 tasks across six domains: eight coding, nine finance, nine policy, nine data analysis, eight education, and seven professional communication tasks. The rubric is also unchanged, with scores assigned for constraint adherence (0 to 4), logical accuracy (0 to 4), and completeness (0 to 2).

The paired within-task design was implemented as specified. Each task was executed under both conditions on both models within $n=1$ per cell, and execution-layer controls ensured that

unpaired data could not be produced inadvertently. The stability subset was also implemented as planned: ten tasks at $n=3$, selected by seeded stratified random draw with quotas weighted toward open-ended domains, frozen prior to any run, and maintained as read-only during execution.

Blinding procedures were maintained and enhanced. Evaluator payloads include only task description, expected constraints, rubric notes, and response text, omitting both model names and condition labels. Output identifiers are opaque sequential tokens, with their mapping stored externally from all evaluator code paths. The family-collision check determines the producing model server-side based on the token, rather than relying on client-side reporting.

CRAFT prompt construction remains unchanged in principle and has been further standardized in practice. Prompts are mechanically assembled from five authored component fields in a fixed order, precluding manual authoring or overrides. This approach eliminates the researcher's degrees of freedom and ensures that the registry can regenerate every prompt in the study. The task stimulus is appended identically to both conditions, in the same position and with the same delimiters, as verified across all 50 tasks. Therefore, the stimulus does not constitute part of either prompt condition.

What was added

Two additional elements were incorporated to strengthen the study design beyond what the original proposal described.

The first is the two-judge architecture. The proposal specified an unbiased LLM judge, and the executed design uses two. Gemini-3.7–flash is the primary model for every output in the study, and a secondary judge rotates by model, so GPT judges Claude output and Claude judges GPT output. No model ever evaluates its own family, and the system enforces this server-side as a hard rejection rather than a convention. Headline results use primary-judge scores only, and the two judges are never averaged, because averaging would collapse their disagreement into a point estimate and destroy the reliability measurement that the second judge exists to produce, and because the secondary varies by producing model, so averaging would bake GPT-as-judge into the Claude distribution and Claude-as-judge into the GPT distribution. Agreement between the two is reported separately instead.

The second enhancement is provenance capture. Neither Anthropic nor Google provides dated snapshot identifiers for the models used, so the identifiers are bare and repointable. To address this, timestamped provider manifests were captured before and after each dispatch period. Every result carries its producing model’s provenance fingerprint, and every evaluation carries its judge’s. A drift check compares live provenance against the most recent manifest and flags any changes.

Known limitations

Commercial models are not archival, and two of the three model identifiers cannot be linked to a fixed version. As a result, the findings are attributable only to the model state observed during the run window, as documented in the captured manifests. This limitation arises from the provider ecosystem rather than from the study design.

Cross-model comparisons are descriptive only, as the secondary varies by producing model and the two models operate under different decoding regimes. The baseline versus CRAFT comparison within each model remains unaffected by these factors.

Evaluator reliability was limited for the primary judge, although no data was lost as a result. Gemini-3.7-flash required retries on 70.9% of its 261 calls and failed outright on 22.2%, with failures concentrated in identifiable degradation windows. All failures were due to 503-exhaustion and were resolved through subsequent evaluation passes, ensuring no loss of scores. During the same period, secondary judges recorded a 10.2% retry rate and 14.4% failure rate for claude-sonnet-5, and zero for both metrics on gpt-5.5. On the generation side, across all 320 runs, there were no retries or failures, indicating that all operational issues were confined to the judge side.

The effect size is small relative to noise. The observed pooled delta of -0.30 exceeds the mean within-cell run-to-run standard deviation of 0.220 by 1.37, so the finding is resolvable only in aggregate. No individual pair's delta is interpretable at $n=1$, and the pooled result holds because 100 paired comparisons average out the noise.

Sixteen secondary evaluations required an extended judge output budget of 12,000 tokens instead of 4,000 after the judge exhausted its budget on reasoning and produced no text. All instances involved Claude-sonnet-5 as the secondary judge on GPT-produced output. Since all headline figures are based solely on primary-judge scores and no primary evaluation ran at a

non-standard budget, this issue affects only inter-rater agreement figures. It influenced which cells qualified as complete but did not impact any headline results.

One of the 320 generations reached the output limit. Task T010, using claude-sonnet-5 in the baseline condition, hit the 4,000-token cap. This instance was retained rather than re-run, as re-running at a higher budget would have introduced different decoding parameters from its paired counterpart. Because truncation occurred in the baseline condition, it reduced baseline completeness points and introduced a bias in favor of CRAFT, representing a conservative direction relative to the reported finding.

The constraint-type analysis presented below is exploratory rather than confirmatory. Only seven of the 50 tasks include three or more negative constraints, and none include more than three; these tasks are also clustered by domain.

Ceiling effect

The baseline condition achieved a score of 9.12 out of 10 when pooled across 100 paired cells, with scores of 9.28 for Claude and 8.96 for GPT. Additionally, 54% of all pairs tied at 10/10. Under these circumstances, the CRAFT condition has minimal headroom, as it matches or underperforms baseline, with little opportunity for further improvement.

Difficulty labels did not correspond to judged difficulty. Tasks labeled as Hard outscored those labeled as Medium at baseline, with scores of 9.20 versus 9.09, and the ceiling effect was consistent across all three tiers. The labels did not reflect the difficulty the models experienced.

This discrepancy is attributable to the task set being calibrated against source material intended for an earlier model generation, but evaluated using 2026 frontier models that solved the tasks with ease- a direct consequence of the model substitutions. The study was designed for one capability level but executed at another. The rubric still registers failures, including cells scoring 3 and below, so the instrument retains discriminatory power, but there is little variation to discriminate at this task difficulty.

Results

All figures presented below are calculated over paired tasks by model cells, using primary-judge scores exclusively. Deltas represent CRAFT minus baseline. An aggregation defect found mid-run had been flattering the CRAFT condition: results were grouped by task alone, pooling the two models' runs as repeat runs of a single task and computing condition means over unequal task sets. Correcting it moved the delta observed at that checkpoint from -0.34 to -0.41, so the correction went against CRAFT rather than toward it. All figures below are computed on the corrected basis across the full set of 100 paired cells.

	claude-sonnet-5	gpt-5.5	pooled
Pairs	50	50	100
Baseline mean	9.28	8.96	9.12
CRAFT mean	8.74	8.90	8.82
Delta	-0.54	-0.06	-0.30

Wins/losses/ties	2/16/32	9/9/32	11/25/64
Wilcoxon p	0.0028	0.5518	0.0225
Paired t	t(49)=-2.944,p=0.004 9	t(49)=-0.305, p=0.7617	t(99)=-2.207, p=0.0297
Cohen's d_z	-0.416	-0.043	-0.221

For Claude-Sonnet-5, CRAFT produced a statistically significant but small-to-medium negative effect. For gpt-5.5, the effect is null across all measures. The pooled significant result averages one real effect and one null effect, so per-model figures serve as the primary reporting unit, with the pooled column providing context. That asymmetry appears in three analyses that do not share a computation path.

Analysis	claude-sonnet-5	gpt-5.5
Mean delta	-0.54	-0.06
Significance	p=0.0028, d_z=-0.42	P = 0.5518, d_z=-0.04
Constraint-type correlation	r= -0.31	r= +0.11

In each analysis, the pooled figure averages one real effect and one null or opposite effect, indicating that the observed asymmetry is not attributable to any single analytical choice. By difficulty tier, the delta is negative in eight of nine cells, with the only positive value being Easy on GPT (+0.33 across six pairs).

What drove the losses

Subscore pattern	Count
Constraint adherence only	8
Constraint + logical	7
Constraint + logical + completeness	6
Completeness only	2
Logical + completeness	1
Logical only	1

Constraint adherence accounts for 21 of the 25 losses and is the sole factor in eight cases. The characteristic loss involves a decrease from 4 to 3 in constraint adherence, with other subscores remaining unchanged. Completeness is seldom implicated, indicating that the penalty is not due to CRAFT verbosity exceeding length limits, but rather to constraint drift.

Mechanism

Two extreme cases, both with perfect agreement between the two independent judges, show the same CRAFT behavior producing opposite outcomes. In T046, a communication task labeled Easy and run on Claude, the constraints require scope discipline, meaning the response should answer the question asked and omit unrelated account matters. CRAFT produced a comprehensive reply raising three additional items and scored 10 down to 3. In T029, a policy

task where Claude judged GPT output, the constraints require alternatives beyond the denial itself. Baseline gave the correct outcome without them, and CRAFT produced structured coverage including four non-waiver options, scoring up from 5 to 10.

The interpretation is that CRAFT consistently elicits comprehensive, structured output. This tendency is advantageous when task constraints reward coverage, but it is penalized when constraints require restraint. Testing this hypothesis against constraint composition supports one model only, and only weakly. For Claude-Sonnet-5, this effect is evident with $r = -0.31$ and a mean delta of -1.57 on tasks with three or more negative constraints, or -0.67 excluding the extreme case T046, and is absent in gpt-5.5 ($r = +0.11$). The pooled null correlation of $r = -0.09$ reflects the average of these opposing patterns rather than an absence of effect.

Negative-constraint load does not distinguish wins from losses in aggregate, with means of 1.45 and 1.56, respectively. Thus, the mechanism is apparent in the Claude-specific slope and in paired case analysis, rather than as a consistent monotonic gradient.

Inter-rater reliability

Metric	Exact agreement	MAD
Constraint adherence (0-4)	75.0%	0.285
Logical accuracy (0-4)	77.5%	0.265
Completeness (0-2)	92.5%	0.095
Total (0-10)	65.0%	0.645

The intraclass correlation coefficient (ICC (3,1)) for total scores was 0.821 (two-way mixed, single measure, consistency). The judges are fixed and identified, so no generalization to a broader population of judges is claimed. Fourteen of 200 cells show disagreements greater than two points, and twelve of these involve the secondary judge scoring lower against a primary score of 9 or 10, which indicates that the primary judge is systematically more generous at the upper end of the scale.

Stability subset

Ten frozen tasks across two conditions and two models at n=3, primary-judge scores only.

Cut	Mean within-cell SD
All 40 cells	0.220
claude-sonnet-5	0.298
gpt-5.5	0.141
Baseline condition	0.227
CRAFT condition	0.212

Thirty-one of the 40 cells exhibited zero variance across all three runs. Two conclusions follow. First, the penalty is reproducible rather than attributable to variance: CRAFT responses are slightly less variable than baseline (0.212 vs. 0.227), and T046 scored 4, 4, and 3 across its three

CRAFT runs, indicating consistent penalization rather than random sampling error. This addresses concerns that the negative delta could result from higher-variance output. Second, the two models differ by approximately a factor of two in run-to-run variance under their respective provider-default decoding regimes (0.298 vs. 0.141), and GPT's variance is identical across conditions to three decimal places, indicating that variance is entirely task-driven. The delta-to-noise ratio is 1.37.

Provenance

Model fingerprints for all three models remained byte-identical across every manifest captured during the approximately 18-hour study window, with captures bracketing each dispatch period. The manifest files confirmed that no model changed during the study.